



## CERTIFICATE OF TESTS

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Printed: 05/16/2024

Heat: 2237327  
Material: 5L- X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)  
Vendor: Nucor Hickman - Armored AR US (180436)

Purchase Order: 26632645  
Sales Order: 1001000690  
MFG Date: 03/04/2024

### CHEMICAL PROPERTIES

|       | C      | MN     | P      | S      | NI     | CR     | CU     | MO     | SI     | TI     | V      | CE     |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Ladle | 0.0500 | 1.0600 | 0.0090 | 0.0014 | 0.0400 | 0.0400 | 0.1200 | 0.0100 | 0.2000 | 0.0140 | 0.0040 | 0.1195 |
|       | Al:    | 0.0300 | Sn:    | 0.0050 | N:     | 0.0060 | B:     | 0.0000 | Ca:    | 0.0010 | Nb:    | 0.0320 |
| Chem  | 0.0559 | 1.1095 | 0.0079 | 0.0014 | 0.0449 | 0.0391 | 0.1039 | 0.0098 | 0.2120 | 0.0129 | 0.0043 | 0.1284 |
|       | Al:    | 0.0381 | Sn:    | 0.0045 | N:     | 0.0804 | B:     | 0.0002 | Ca:    | 0.0018 | Nb:    | 0.0330 |
| Chem  | 0.0572 | 1.1126 | 0.0078 | 0.0011 | 0.0447 | 0.0388 | 0.1029 | 0.0095 | 0.2100 | 0.0129 | 0.0041 | 0.1294 |
|       | Al:    | 0.0376 | Sn:    | 0.0044 | N:     | 0.0276 | B:     | 0.0001 | Ca:    | 0.0018 | Nb:    | 0.0322 |

Acceptance Criteria: CEPCM = 0.2500

### MECHANICAL PROPERTIES

YS (ksi): 73.90 TS (ksi): 77.80 EL (%): 28.00 HRB: 89.00 Y/T(%): 0.95

### CHARPY TEST RESULTS (BODY)

| Orientation : TRANSVERSE             | Specimen(S) | Energy (ft-lb) | %Shear |
|--------------------------------------|-------------|----------------|--------|
| Spec. Size (in) : .394"X.197"X2.165" | 1           | 81.00          | 100.00 |
| Test Temp (°F) : -20                 | 2           | 77.00          | 100.00 |
| Location : 3T90                      | 3           | 76.00          | 100.00 |
|                                      | Average     | 78.00          | 100.00 |

Acceptance Criteria Min. (ft-lb) : 10

### COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION  
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT  
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM  
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE  
FLATTENING TEST PASSED  
1 INCH L90 STRIP SPECIMEN  
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION  
API 5L PSL 2 GRADE X52M  
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1  
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION  
MELTED AND MANUFACTURED IN THE U.S.A.  
STEEL SUPPLIER: Nucor Hickman - Armored AR US  
COIL MANUFACTURER: Nucor Hickman - Armored AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept